

Building instruction

Step 1:

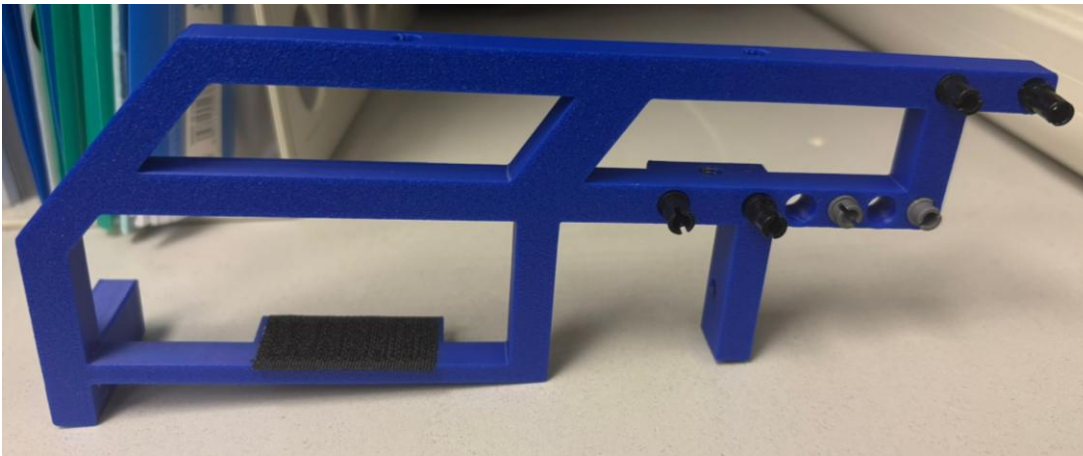


Figure 1 : Step 1 Right-side finishing frame

First, plug 4 Lego connector peg with friction, 2-module and 2 Lego 1 1/2 connecting bush into the holes shown in the above figure. Do the same thing to the left sided car frame.

Step 2 :



Figure 2: Step 2 finishing figure

Then, put 2 frame together and connector with 3 Lego technic pin connector block 1x3x3 module like the figure shown.

Step 3 :

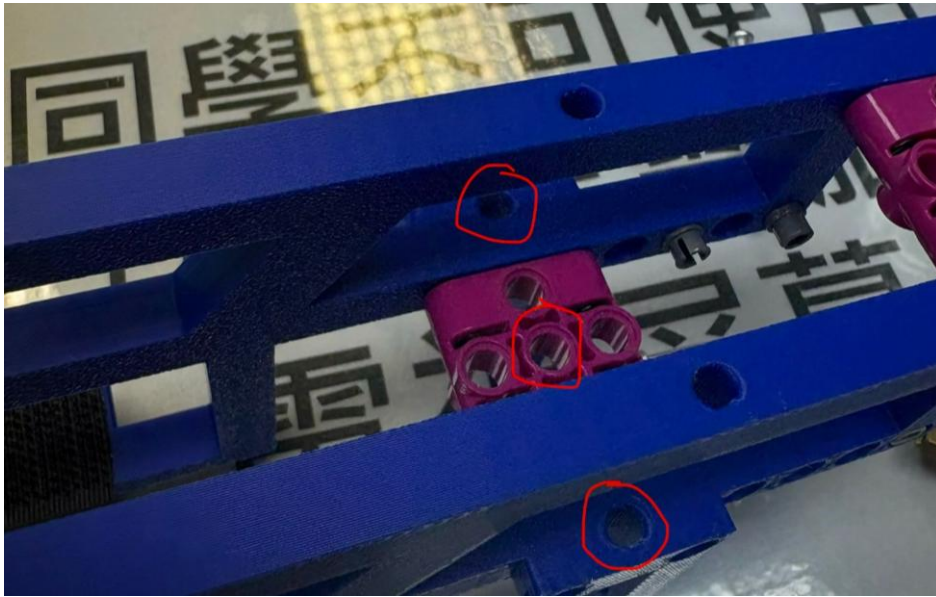


Figure 3 : Figure for locking the motor tips

3 M4 socket head screws are placed in the holes drawn with red circles, showing in the above figure. They are used for fixing up the moving motor.

Step 4 :

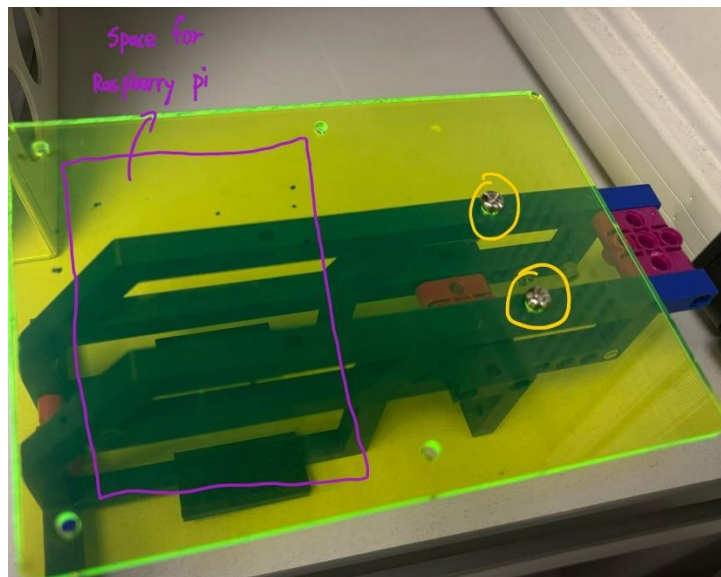


Figure 4 : Space for Raspberry Pi (Purple rectangle), M4x15 socket screws (yellow circles)

Next, place the laser cut middle plate on the top of 2 side car frame and then use 2 M4 x15 socket head screw and 2 lock nuts to fix the plate and frame as shown as the hole with yellow circles in the above figure.

Step 5:

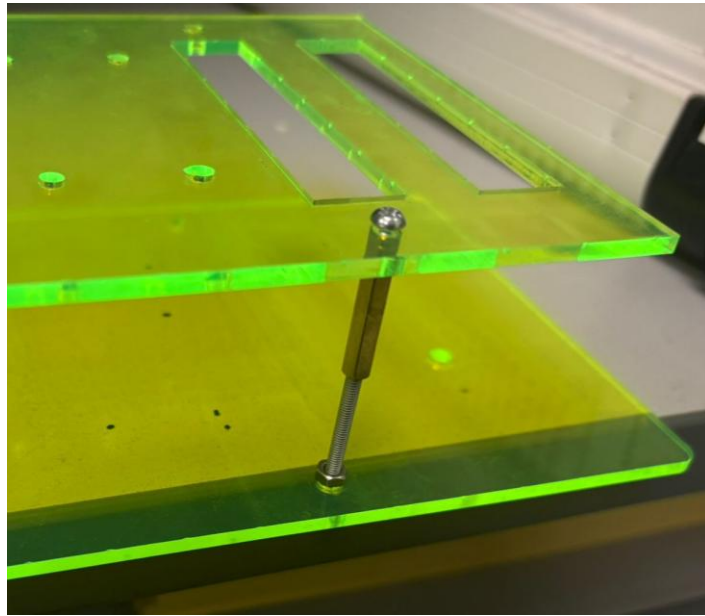


Figure 5 : finishing step 5

Next, use 1 M3 x 40 ,1 M3 x 15 socket head screw and 1 M3 Double Through Brass Standoffs Screws Hex Spacer Standoffs to fix the laser cut middle plate and the top plate. M3 x 40 passes through the hole on the middle plate and M3 x 15 socket head screw passes through the top plates. Two screws are connector by the hex spacer standoffs.

Step 6 :

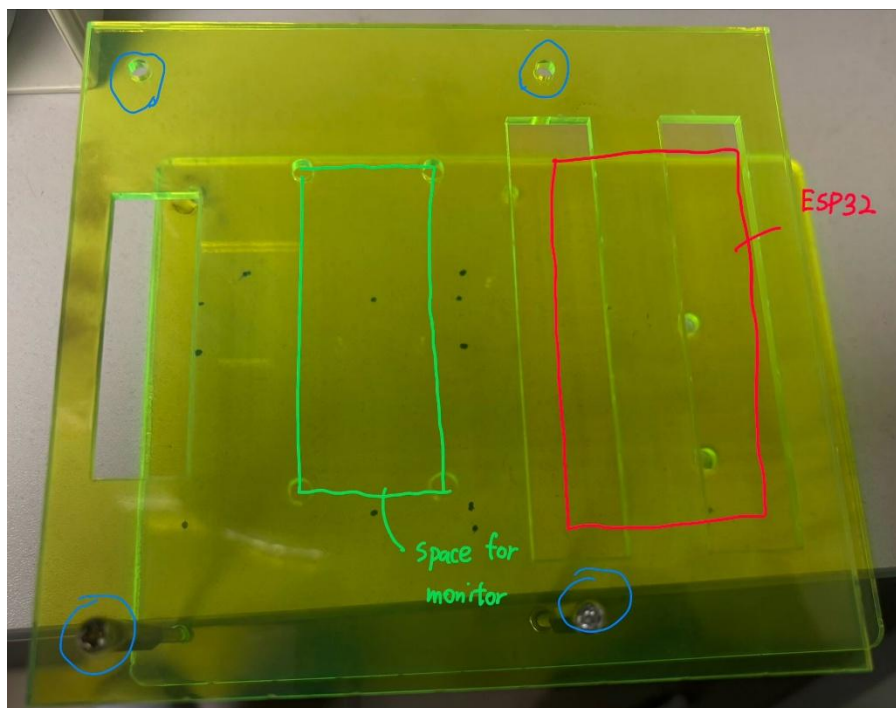


Figure 6 : Hints for placing monitor and ESP32

Last, repeat step 5 for 3 times for other 3 holes circling with blue color in the figure shown above. Place the monitor

(green rectangle box) and fix with 4 M3 socket head screws.
For the ESP32, place at the space drawing in red rectangle box.

Tips for fixing ESP32 ,Raspberry Pi and the battery box

All these 3 electronic components are fixed by hook and loop fasteners.
Demonstration diagrams are shown below.

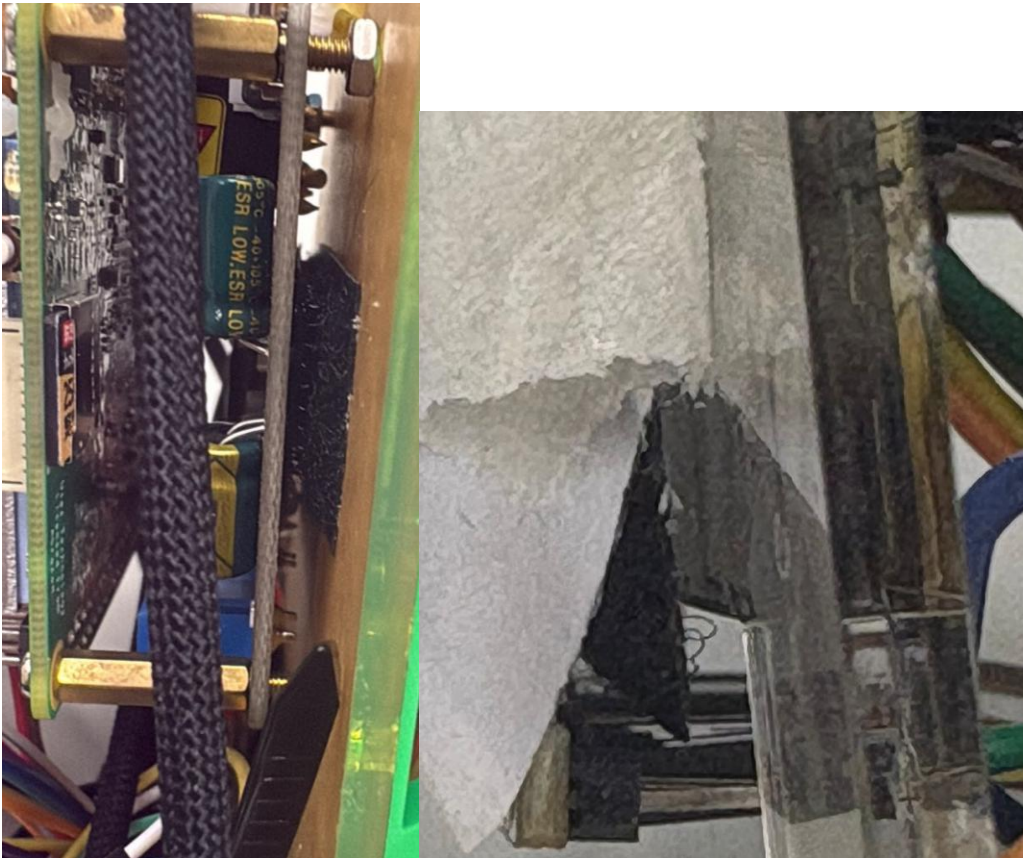


Figure 7&8 : Demonstration of the hook and loop fasteners

For the differential part, it is fixed on the steering servo and also connector with the moving motor.

